Inequalities — problem 7.28

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A point is chosen on each side of a unit square. The four points are sides of a quadrilateral with sides a, b, c, d. Show that

$$2 \leq a^2 + b^2 + c^2 + d^2 \leq 4 \quad \text{and} \quad 2\sqrt{2} \leq a + b + c + d \leq 4.$$